ADVENTIST UNIVERSITY OF CENTRAL AFRICA (AUCA)

Master of Science in Data Analytics

MSDA9215 - Big Data Analytics · Final Project, October 2026

Distributed Multi-Model Analytics for E-commerce Data

Appendices to the Technical Report

A: How to reproduce · B: Key scripts · C: Evidence screenshots S1-S14

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Appendix A - How to reproduce

Everything runs in Docker. From Windows PowerShell, in the project folder:

```
docker compose up -d --build                    # mongo :27017, hbase :9090 / UI :16010, app (Spark UI :4040)
docker compose exec -T app python wait_for_services.py
.\run_all.ps1                                   # whole pipeline (or .\run_all.ps1 -Scale 1.0)
.\evidence.ps1                                  # re-creates the evidence screenshots step by step
```

Step	Command (prefix each with docker compose exec -T app)	Output
1 Generate data	`python data_generation/dataset_generator.py --out data/raw --scale 0.1`	data/raw/*.json
2 MongoDB	`python mongodb/load_mongodb.py --sessions-limit 50000`; `python mongodb/aggregations.py`	5 collections, indexes, output/mongo_*.csv
3 HBase	`python hbase/load_hbase.py --recreate`; `python hbase/query_hbase.py`	2 tables; queries Q1-Q5
4 Spark clean	`python spark/01_clean_normalize.py`	data/clean/*.parquet, data_quality_report.csv
5 Spark analytics	`python spark/02_recommendations.py`; `03_cohort_analysis.py`; `04_spark_sql_analytics.py`	output/reco_, *cohort_*, sql_*.csv
6 Integration	`python integration/clv_analysis.py --mongo mongo --hbase hbase`; `python integration/funnel_analysis.py --mongo mongo --hbase hbase`	output/clv_*.csv, funnel_*.csv
7 Charts	`python visualization/make_charts.py`	report/figures/*.png
Shells	`Get-Content mongodb\evidence.js | docker exec -i mongo mongosh ecommerce --quiet`; `Get-Content hbase\evidence.hbase | docker exec -i hbase hbase shell -n`	S7, S8

Repository path	Purpose
data_generation/dataset_generator.py	Improved generator (original kept as dataset_generator_original.py)
mongodb/load_mongodb.py, aggregations.py, aggregations.js, evidence.js	Document model, bulk load, indexes, 4 pipelines
hbase/create_tables.hbase, load_hbase.py, query_hbase.py, sample_queries.hbase, hbase_client.py	Schema, loader, queries, Thrift client + local emulator
spark/spark_utils.py, 01-04 *.py	Explicit schemas, cleaning, recommendations, cohorts, Spark SQL
integration/sources.py, clv_analysis.py, funnel_analysis.py	Source-independent loaders (range scans) and the two integrated queries
visualization/make_charts.py, report/make_architecture_diagram.py	Charts and architecture diagram
Dockerfile, docker-compose.yml, run_all.ps1, run_all.sh, evidence.ps1, wait_for_services.py, README.md	Docker environment, orchestration, documentation

Appendix B - Key scripts

B.1 HBase schema creation (hbase/create_tables.hbase)

```
# create_tables.hbase - HBase schema (run with:  hbase shell hbase/create_tables.hbase
#                                    or docker exec -i hbase hbase shell < hbase/create_tables.hbase)
#
# ------------------------------------------------------------------------
# TABLE 1: user_sessions  - time-series browsing activity per user
# ------------------------------------------------------------------------
# Row key  : <user_id>|<reverse_ts>|<session_id>
#            reverse_ts = 9999999999999 - start_time_epoch_millis (13 digits, zero padded)
#            -> all sessions of a user are contiguous (one prefix scan)
#            -> newest session sorts FIRST (no client-side sort for "recent activity")
#            -> session_id suffix keeps keys unique if two sessions start in the same ms
#
# Column families (kept few and small, each stored in its own HFiles):
#   info : session header - one cell per attribute (start, end, duration, device, geo, referrer,
#          conversion_status, n_pages ...). Read by "list sessions" queries -> smallest I/O.
#   pv   : page views, one column per view: qualifier = 3-digit sequence ("000","001",...),
#          value = "page_type|product_id|category_id|view_duration|timestamp".
#          Sparse & variable width: a 3-page session stores 3 cells, a 15-page session 15.
#   cart : one column per product in the cart: qualifier = product_id, value = "qty|unit_price".
#          Most sessions have 0-2 cells here -> sparse storage costs nothing for empty cells.
#
# VERSIONS => 1  : events are immutable, no need for history
# COMPRESSION    : SNAPPY (falls back to NONE if native libs are missing; change if needed)
# BLOOMFILTER ROW: speeds up point gets of a single session row
# TTL on pv      : 180 days of raw click-stream (older detail can be archived to HDFS/parquet)
# SPLITS         : pre-split on user-id ranges so the bulk load is spread over regions
create 'user_sessions',
  {NAME => 'info', VERSIONS => 1, COMPRESSION => 'NONE', BLOOMFILTER => 'ROW'},
  {NAME => 'pv',   VERSIONS => 1, COMPRESSION => 'NONE', TTL => 15552000},
  {NAME => 'cart', VERSIONS => 1, COMPRESSION => 'NONE'},
  SPLITS => ['user_002', 'user_004', 'user_006', 'user_008']


# ------------------------------------------------------------------------
# TABLE 2: product_metrics - daily performance per product
# ------------------------------------------------------------------------
# Row key  : <product_id>|<yyyymmdd>
#            -> a product's whole history is one contiguous range;
#               a date range is a bounded scan: STARTROW prod_x|20260901, STOPROW prod_x|20261001
# Column family m (metrics): views, cart_adds, purchases, units_sold, revenue_cents
#   Batch loads write pre-aggregated values; a streaming job would use atomic counters:
#   incr 'product_metrics', 'prod_00123|20261015', 'm:views', 1
create 'product_metrics',
  {NAME => 'm', VERSIONS => 1, COMPRESSION => 'NONE', BLOOMFILTER => 'ROW'},
  SPLITS => ['prod_01', 'prod_02', 'prod_03', 'prod_04']


list
describe 'user_sessions'
describe 'product_metrics'
```

B.2 HBase sample queries (hbase/sample_queries.hbase)

```
# sample_queries.hbase - HBase shell equivalents of query_hbase.py
#   hbase shell hbase/sample_queries.hbase
# Row keys: user_sessions   = user_id|reverse_ts|session_id   (reverse_ts = 9999999999999 - epoch_ms)
#           product_metrics = product_id|yyyymmdd


# Q1 - the 5 most recent sessions of a user (newest first thanks to the reversed timestamp),
#      reading only the small 'info' family
scan 'user_sessions', {ROWPREFIXFILTER => 'user_000042|', COLUMNS => ['info'], LIMIT => 5}


# Q2 - a user's sessions that started in September 2026.
#      newer bound -> STARTROW, older bound -> STOPROW  (because the timestamp is reversed)
#      reverse(2026-10-01T00:00:00Z - 1 ms) = 8209187200000 ; reverse(2026-09-01T00:00:00Z) = 8211779199999
scan 'user_sessions', {STARTROW => 'user_000042|8209187200000', STOPROW => 'user_000042|8211779199999~',
                       COLUMNS => ['info:start_time', 'info:conversion_status']}


# Q3 - complete click-stream (page views + cart) of a single session (point get by full row key;
#      take a key returned by Q1)
get 'user_sessions', 'user_000042|8206766047273|sess_2817bd6893', {COLUMN => ['pv', 'cart']}


# Q3b - only the converted sessions of a user (server-side filter on a column value)
scan 'user_sessions', {ROWPREFIXFILTER => 'user_000042|',
  FILTER => "SingleColumnValueFilter('info','conversion_status',=,'binary:converted',true,true)"}


# Q4 - number of sessions of a user (counts rows under the prefix)
count 'user_sessions', {ROWPREFIXFILTER => 'user_000042|', INTERVAL => 1000}


# Q5 - a product's daily performance during September 2026 (contiguous range scan)
scan 'product_metrics', {STARTROW => 'prod_00123|20260901', STOPROW => 'prod_00123|20261001'}


# Q5b - one metric for one day
get 'product_metrics', 'prod_00123|20260915', 'm:views'


# Streaming-style update: atomic counter increments (what a real-time click consumer would do)
incr 'product_metrics', 'prod_00123|20261101', 'm:views', 1
```

```
get_counter 'product_metrics', 'prod_00123|20261101', 'm:views'
```

B.3 MongoDB aggregation pipelines (mongodb/aggregations.js)

```
// aggregations.js - the same analytics pipelines as aggregations.py, for mongosh.
// Run:  mongosh "mongodb://localhost:27017/ecommerce" mongodb/aggregations.js
// (or docker exec -i mongo mongosh ecommerce < mongodb/aggregations.js)


// ---------------------------------------------------------------- 1. Product popularity
print("\n=== 1. Top-selling products ===");
printjson(db.transactions.aggregate([
  { $match: { status: { $in: ["completed", "shipped", "delivered", "processing"] } } },
  { $unwind: "$items" },
  { $group: { _id: "$items.product_id", product_name: { $first: "$items.product_name" },
              category_id: { $first: "$items.category_id" },
              units_sold: { $sum: "$items.quantity" }, revenue: { $sum: "$items.subtotal" },
              orders: { $sum: 1 } } },
  { $sort: { units_sold: -1, revenue: -1 } },
  { $limit: 10 },
  { $lookup: { from: "products", localField: "_id", foreignField: "_id", as: "product" } },
  { $unwind: "$product" },
  { $project: { _id: 0, product_id: "$_id", product_name: 1, category_id: 1, units_sold: 1, orders: 1,
                revenue: { $round: ["$revenue", 2] }, current_stock: "$product.current_stock" } }
]).toArray());


// ---------------------------------------------------------------- 2. Revenue by category & month
print("\n=== 2. Revenue & estimated profit by category and month ===");
printjson(db.transactions.aggregate([
  { $unwind: "$items" },
  { $lookup: { from: "categories", localField: "items.category_id", foreignField: "_id", as: "cat" } },
  { $unwind: "$cat" },
  { $addFields: { sub: { $arrayElemAt: [ { $filter: { input: "$cat.subcategories", as: "s",
        cond: { $eq: ["$$s.subcategory_id", "$items.subcategory_id"] } } }, 0 ] } } },
  { $group: { _id: { category: "$cat.name", month: { $dateToString: { format: "%Y-%m", date: "$timestamp" } } },
              revenue: { $sum: "$items.subtotal" },
              est_profit: { $sum: { $multiply: ["$items.subtotal", "$sub.profit_margin"] } },
              units: { $sum: "$items.quantity" } } },
  { $sort: { "_id.month": 1, revenue: -1 } },
  { $limit: 30 }
], { allowDiskUse: true }).toArray());

// Same idea restricted to a time window (uses the {timestamp:1} index)
print("\n=== 2b. Revenue in October 2026 ===");
printjson(db.transactions.aggregate([
  { $match: { timestamp: { $gte: ISODate("2026-10-01"), $lt: ISODate("2026-11-01") } } },
  { $group: { _id: null, revenue: { $sum: "$total" }, orders: { $sum: 1 }, aov: { $avg: "$total" } } }
]).toArray());


// ---------------------------------------------------------------- 3. User segmentation
print("\n=== 3. User segmentation by purchase frequency and spend ===");
printjson(db.transactions.aggregate([
  { $group: { _id: "$user_id", orders: { $sum: 1 }, spent: { $sum: "$total" }, last_order: { $max: "$timestamp" } } },
  { $facet: {
      by_frequency: [ { $bucket: { groupBy: "$orders", boundaries: [1, 3, 6, 10, 1000], default: "other",
                        output: { users: { $sum: 1 }, avg_spent: { $avg: "$spent" }, total_revenue: { $sum:
"$spent" } } } } ],
      by_spend: [ { $bucket: { groupBy: "$spent", boundaries: [0, 500, 1500, 3000, 6000, 1e9], default: "other",
                    output: { users: { $sum: 1 }, avg_orders: { $avg: "$orders" } } } } ],
      top_customers: [ { $sort: { spent: -1 } }, { $limit: 5 } ]
  } }
], { allowDiskUse: true }).toArray());


// Because users embed purchase_summary, the "top customers" question is a simple indexed find:
print("\n=== 3b. Top customers straight from the users collection ===");
printjson(db.users.find({}, { "purchase_summary.total_spent": 1, "purchase_summary.order_count": 1, geo_data: 1 })
  .sort({ "purchase_summary.total_spent": -1 }).limit(5).toArray());


// ---------------------------------------------------------------- 4. Journey funnel (sessions sample)
print("\n=== 4. Session funnel by referrer ===");
printjson(db.sessions.aggregate([
  { $project: { referrer: 1, conversion_status: 1, pages: "$page_views.page_type" } },
  { $group: { _id: "$referrer", sessions: { $sum: 1 },
      viewed_product: { $sum: { $cond: [{ $in: ["product_detail", "$pages"] }, 1, 0] } },
      reached_cart: { $sum: { $cond: [{ $in: ["cart", "$pages"] }, 1, 0] } },
      reached_checkout: { $sum: { $cond: [{ $in: ["checkout", "$pages"] }, 1, 0] } },
      converted: { $sum: { $cond: [{ $eq: ["$conversion_status", "converted"] }, 1, 0] } } } },
  { $addFields: { conversion_rate: { $round: [{ $divide: ["$converted", "$sessions"] }, 4] } } },
  { $sort: { conversion_rate: -1 } }
]).toArray());


// Full journey of one user's most recent session - one document read, no joins
print("\n=== 5. Latest session journey for user_000042 ===");
printjson(db.sessions.find({ user_id: "user_000042" }).sort({ start_time: -1 }).limit(1).toArray());
```

B.4 Spark SQL over MongoDB-shaped and HBase-shaped data (excerpt)

```
WITH engagement AS (                                    -- HBase: user_sessions rows
    SELECT SPLIT(rowkey, '\\|')[0]              AS user_id,
           COUNT(*)                              AS sessions,
           ROUND(AVG(`info:duration`) / 60, 1)   AS avg_session_min
    FROM hbase_user_sessions GROUP BY 1
)
```

```
SELECT CASE WHEN e.sessions >= 25 THEN '25+ sessions' WHEN e.sessions >= 20 THEN '20-24'
            WHEN e.sessions >= 15 THEN '15-19' ELSE '<15' END    AS engagement_band,
       COUNT(*)                                                  AS users,
       ROUND(AVG(m.purchase_summary.order_count), 2) AS avg_orders,       -- MongoDB: users document
       ROUND(AVG(m.purchase_summary.total_spent), 2) AS avg_spent
FROM mongo_users m JOIN engagement e ON m._id = e.user_id
GROUP BY 1 ORDER BY 1;
```

B.5 CLV core computation (integration/clv_analysis.py, excerpt)

```
purchases = (items_m.groupBy("user_id").agg(
    F.countDistinct("transaction_id").alias("orders"),
    F.round(F.sum("net_revenue"), 2).alias("revenue"),
    F.round(F.sum("profit"), 2).alias("gross_profit"),
    F.max("txn_time").alias("last_purchase"))
    .withColumn("aov", F.col("revenue") / F.col("orders"))
    .withColumn("margin_pct", F.col("gross_profit") / F.col("revenue"))
    .withColumn("orders_per_month", F.col("orders") / F.lit(window_months)))


lifetime_months = min(1 / (1 - retention), 36)          # retention estimated from the data
clv = clv.withColumn("clv", F.col("aov") * F.col("orders_per_month")
                            * F.col("margin_pct") * F.lit(lifetime_months))
```

B.6 Robust HBase reader: row-key range scans with retry (integration/sources.py)

```
def _scan_in_ranges(source, table, columns, prefixes, batch_size=200, retries=4):
    """Scan an HBase table as many short row-key RANGE scans instead of one full-table scan.


    A single full scan through the Thrift gateway keeps one scanner open for minutes, which a
    single-node HBase in Docker can stall on (TimeoutError). Short range scans - the access path
    the row-key design is built for - finish quickly, and a failed range is simply retried on a
    fresh connection, so the job is robust. Yields (rowkey, cells).
    """
    import time
    conn = _hbase(source)
    for i, start in enumerate(prefixes):
        stop = prefixes[i + 1] if i + 1 < len(prefixes) else None
        for attempt in range(1, retries + 1):
            try:
                rows = list(conn.table(table).scan(row_start=start.encode(),
                                                   row_stop=stop.encode() if stop else None,
                                                   columns=columns, batch_size=batch_size))
                break
            except Exception as e:  # noqa: BLE001 - Thrift timeout / broken pipe
                if attempt == retries:
                    raise
                print(f"   range {start}: {e.__class__.__name__}, retry {attempt}/{retries - 1}")
                time.sleep(3 * attempt)
                try:
                    conn.close()
                except Exception:  # noqa: BLE001
                    pass
                conn = _hbase(source)
        for row in rows:
            yield row
        if (i + 1) % 20 == 0:
            print(f"   {table}: {i + 1}/{len(prefixes)} key ranges scanned")



# user ids are user_000000..user_009999 -> 100 ranges of 100 users; products prod_00000..prod_04999
USER_RANGES = [f"user_{i:04d}" for i in range(100)]
PRODUCT_RANGES = [f"prod_{i:03d}" for i in range(50)]
```

B.7 Docker environment (docker-compose.yml)

```
# Whole project in Docker:  docker compose up -d --build
#   mongo  - MongoDB 7                      (localhost:27017)
#   hbase  - HBase standalone + Thrift      (UI http://localhost:16010, Thrift 9090)
#   app    - Python 3.11 + Java 17 + PySpark (runs every script; Spark UI http://localhost:4040)
services:
  mongo:
    image: mongo:7
    container_name: mongo
    ports:
      - "27017:27017"
    volumes:
      - mongo_data:/data/db


  hbase:
    image: dajobe/hbase
    container_name: hbase
    hostname: hbase
    ports:
      - "16010:16010"   # HMaster web UI
      - "9090:9090"     # Thrift API (used by happybase)
      - "9095:9095"     # Thrift web UI
      - "2181:2181"     # ZooKeeper
    volumes:
      - hbase_data:/data


  app:
    build: .
    container_name: app
```

```
    depends_on:
      - mongo
      - hbase
    environment:
      MONGO_URI: mongodb://mongo:27017
      HBASE_HOST: hbase
      HBASE_PORT: "9090"
      SPARK_DRIVER_MEMORY: 3g
    ports:
      - "4040:4040"     # Spark UI while a job is running
    volumes:
      - .:/app          # your project folder; results land in output/ and report/figures/ on Windows
    working_dir: /app
    command: sleep infinity


volumes:
  mongo_data:
  hbase_data:
```

Appendix C - Evidence screenshots (live Docker run)

C.1 Environment

S1 - Running containers

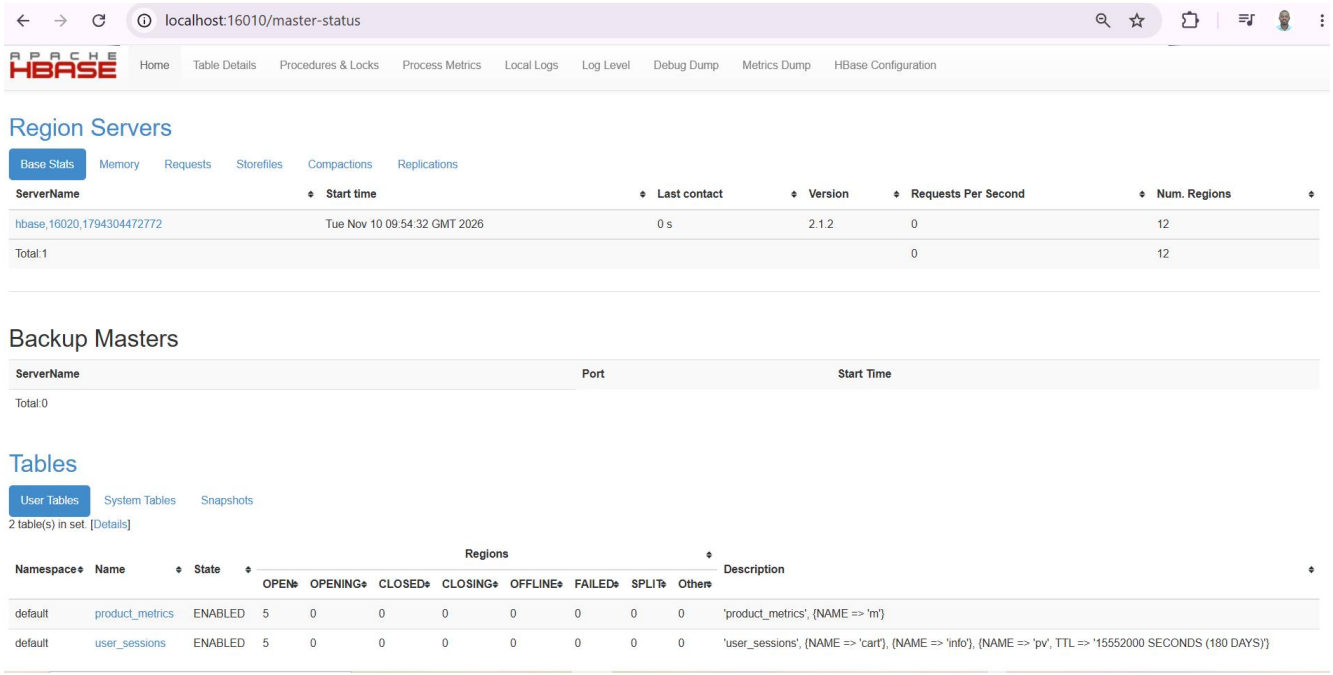
Command: `docker compose ps`

```
PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics> docker compose ps
NAME      IMAGE                      COMMAND                  SERVICE   CREATED        STATUS          PORTS
app       e-commerce-analytics-app   "sleep infinity"         app       2 hours ago    Up 2 hours      0.0.0.0:4040->4040/tcp, [::]:4040->4040/tcp
hbase     dajobe/hbase               "/opt/hbase-server"      hbase     21 hours ago   Up 47 minutes   0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp, 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp, 0.0.0.0:9095->9095/tcp, [::]:9095->9095/tcp, 0.0.0.0:16010->16010/tcp, [::]:16010->16010/tcp
mongo     mongo:7                    "docker-entrypoint.s…"   mongo     2 hours ago    Up 2 hours      0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics>
```

Comment: The three project containers - app (Python 3.11 + Java 17 + PySpark), hbase (HBase + Thrift) and mongo (MongoDB 7) - are up, with ports 4040, 9090/16010/2181 and 27017 published.

S2 - HBase master web UI

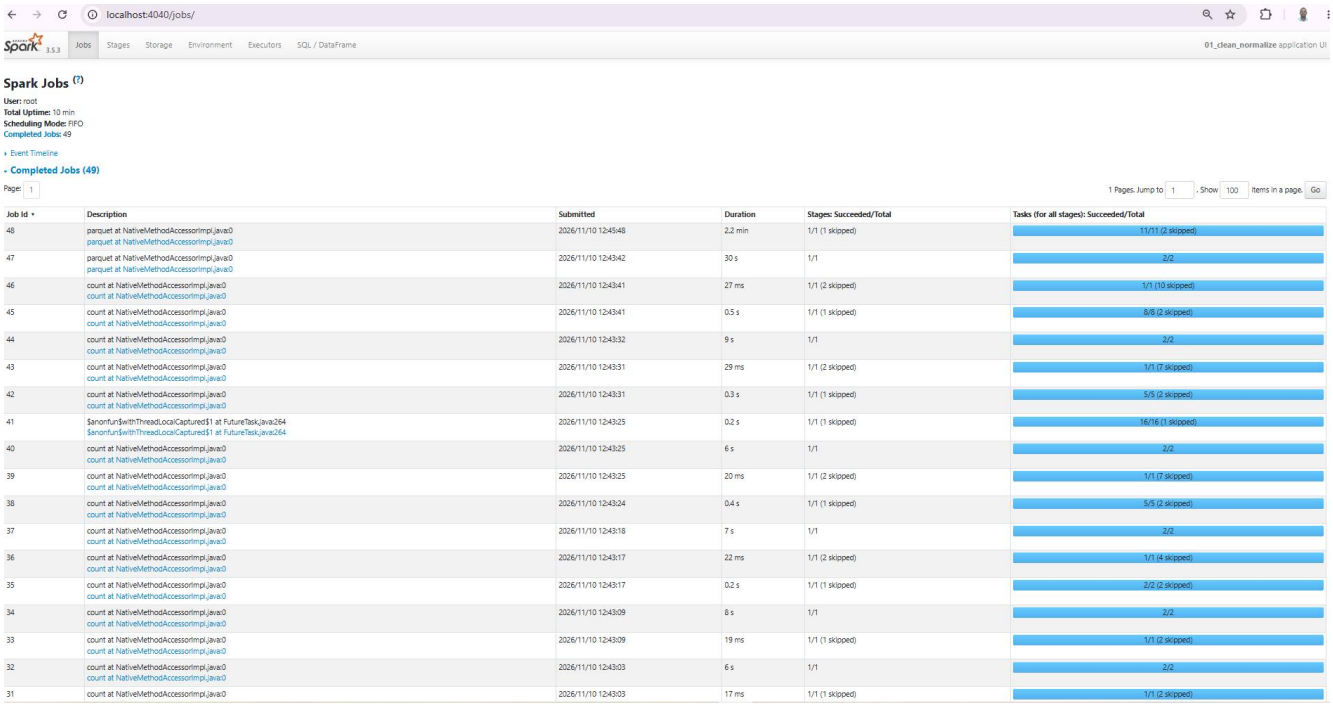
URL: `http://localhost:16010`



Comment: HBase 2.1.2 with one region server. Both user tables are ENABLED with **5 regions each**, proving the 4 pre-split points took effect; the description lists the info, pv (TTL 180 days) and cart families.

S3 - Spark UI during the cleaning job

URL: `http://localhost:4040`



Spark Jobs (?)

User: root
Total Uptime: 10 min
Scheduling Mode: FIFO
Completed Jobs: 49

▸ Event Timeline

▾ Completed Jobs (49)

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Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
48	parquet at NativeMethodAccessorImpl.java:0 parquet at NativeMethodAccessorImpl.java:0	2026/11/10 12:45:48	2.2 min	1/1 (1 skipped)	11/11 (2 skipped)
47	parquet at NativeMethodAccessorImpl.java:0 parquet at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:42	30 s	1/1	2/2
46	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:41	27 ms	1/1 (2 skipped)	1/1 (10 skipped)
45	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:41	0.5 s	1/1 (1 skipped)	8/8 (2 skipped)
44	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:32	9 s	1/1	2/2
43	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:31	29 ms	1/1 (2 skipped)	1/1 (7 skipped)
42	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:31	0.3 s	1/1 (1 skipped)	5/5 (2 skipped)
41	$anonfun$withThreadLocalCaptured$1 at FutureTask.java:264 $anonfun$withThreadLocalCaptured$1 at FutureTask.java:264	2026/11/10 12:43:25	0.2 s	1/1 (1 skipped)	16/16 (1 skipped)
40	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:25	6 s	1/1	2/2
39	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:25	20 ms	1/1 (2 skipped)	1/1 (7 skipped)
38	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:24	0.4 s	1/1 (1 skipped)	5/5 (2 skipped)
37	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:18	7 s	1/1	2/2
36	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:17	22 ms	1/1 (2 skipped)	1/1 (4 skipped)
35	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:17	0.2 s	1/1 (1 skipped)	2/2 (2 skipped)
34	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:09	8 s	1/1	2/2
33	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:09	19 ms	1/1 (1 skipped)	1/1 (2 skipped)
32	count at NativeMethodAccessorImpl.java:0 count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:03	6 s	1/1	2/2
31	count at NativeMethodAccessorImpl.java:0	2026/11/10 12:43:03	17 ms	1/1 (1 skipped)	1/1 (2 skipped)

> **Comment:** Spark 3.5.3 ran 49 jobs for 01_clean_normalize: Parquet writes (up to 2.2 min) and the short data-quality counts; skipped stages show cached data being reused.

C.2 Data generation

S4 - Dataset generation, scale 0.1

Command: `python data_generation/dataset_generator.py --out data/raw --scale 0.1`

```
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics>  python data_generation/dataset_generator.py --out data/raw --scale 0.1
Initializing dataset generation (scale=0.1, 200,000 sessions, 50,000 transactions)...
Generated 25 categories
Generated 5000 products
Generated 10000 users
Generating sessions and transactions...
Progress: 50,000/200,000 sessions, 20,260/50,000 transactions (14s)
Progress: 100,000/200,000 sessions, 40,562/50,000 transactions (35s)
Progress: 150,000/200,000 sessions, 50,000/50,000 transactions (49s)
Progress: 200,000/200,000 sessions, 50,000/50,000 transactions (67s)
Saving datasets...

Dataset generation complete in 71s -> data/raw
- Sessions: 200,000 (target: 200,000) in 2 file(s)
- Transactions: 50,000 (target: 50,000)
- Remaining stock units: 2,310,878
```

> **Comment:** 200,000 sessions and 50,000 transactions in 2 files in 71 s. The transaction target is reached between 100,000 and 150,000 sessions; after that converted sessions create no orders - the source of the 15,552 gap.

C.3 Part 1 - MongoDB

S5 - Loading MongoDB

Command: `python mongodb/load_mongodb.py --sessions-limit 50000`

```
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics> python mongodb/load_mongodb.py --sessions-limit 50000
Loading into mongodb://localhost:27017/ecommerce
  categories          25 docs  (0.1s)
  products         5,000 docs  (0.4s)
  transactions    50,000 docs  (3.0s)
  users           10,000 docs  (0.4s)
  sessions        50,000 docs  (5.5s)
  indexes created
Done.
```

> **Comment:** Five collections loaded with unordered batched inserts in ~10 s, then the indexes were created.

S6 - The four aggregation pipelines

Command: `python mongodb/aggregations.py`

```
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics> python mongodb/aggregations.py

== 1. Top-selling products ===
  {'product_name': 'Right-Sized Bifurcated Help-Desk', 'category_id': 'cat_021', 'units_sold': 84, 'orders': 35, 'product_id': 'prod_03248', 'revenue': 31485.72, 'current_price': 374.83, 'current_stock': 816}
  {'product_name': 'Cloned Discrete Array', 'category_id': 'cat_024', 'units_sold': 84, 'orders': 36, 'product_id': 'prod_04028', 'revenue': 12798.24, 'current_price': 152.36, 'current_stock': 241}
  {'product_name': 'Centralized Executive Customer Loyalty', 'category_id': 'cat_018', 'units_sold': 82, 'orders': 35, 'product_id': 'prod_01324', 'revenue': 15967.04, 'current_price': 194.72, 'current_stock': 589}
  {'product_name': 'Optional Solution-Oriented Concept', 'category_id': 'cat_010', 'units_sold': 81, 'orders': 36, 'product_id': 'prod_00810', 'revenue': 30923.37, 'current_price': 381.77, 'current_stock': 918}
  {'product_name': 'Inverse Stable Paradigm', 'category_id': 'cat_018', 'units_sold': 81, 'orders': 38, 'product_id': 'prod_04620', 'revenue': 22634.64, 'current_price': 279.44, 'current_stock': 724}
  {'product_name': 'Universal Methodical Synergy', 'category_id': 'cat_001', 'units_sold': 81, 'orders': 37, 'product_id': 'prod_03004', 'revenue': 19151.64, 'current_price': 236.44, 'current_stock': 877}
  {'product_name': 'Networked Next Generation Capability', 'category_id': 'cat_002', 'units_sold': 81, 'orders': 40, 'product_id': 'prod_00366', 'revenue': 12248.01, 'current_price': 151.21, 'current_stock': 659}
  {'product_name': 'Phased Well-Modulated Encoding', 'category_id': 'cat_005', 'units_sold': 80, 'orders': 35, 'product_id': 'prod_04258', 'revenue': 516.0, 'current_price': 6.45, 'current_stock': 478}
  {'product_name': 'Balanced Foreground Internet Solution', 'category_id': 'cat_023', 'units_sold': 79, 'orders': 37, 'product_id': 'prod_04961', 'revenue': 29087.8, 'current_price': 368.2, 'current_stock': 48}
  {'product_name': 'Re-Contextualized Scalable Access', 'category_id': 'cat_022', 'units_sold': 79, 'orders': 37, 'product_id': 'prod_03342', 'revenue': 28309.65, 'current_price': 358.35, 'current_stock': 288}

== 2. Revenue by category & month (first rows) ===
  {'units': 1927, 'category': 'James LLC', 'month': '2026-08', 'revenue': 552381.46, 'est_profit': 155778.37}
  {'units': 2195, 'category': 'Cameron-Parsons', 'month': '2026-08', 'revenue': 551274.18, 'est_profit': 153095.02}
  {'units': 2108, 'category': 'Moore, Becker and Carlson', 'month': '2026-08', 'revenue': 550233.02, 'est_profit': 148425.05}
  {'units': 2042, 'category': 'Skinner-Romero', 'month': '2026-08', 'revenue': 548425.33, 'est_profit': 141020.27}
  {'units': 2105, 'category': 'Rodriguez-Graham', 'month': '2026-08', 'revenue': 542942.26, 'est_profit': 196341.59}
  {'units': 2046, 'category': 'Patterson, Smith and Jones', 'month': '2026-08', 'revenue': 533810.26, 'est_profit': 148947.54}
  {'units': 2116, 'category': 'Hancock Inc', 'month': '2026-08', 'revenue': 518942.79, 'est_profit': 146813.06}
  {'units': 2099, 'category': 'Mitchell-Kim', 'month': '2026-08', 'revenue': 518068.45, 'est_profit': 115916.68}
  {'units': 2094, 'category': 'Anderson, Walls and Duncan', 'month': '2026-08', 'revenue': 508959.42, 'est_profit': 132037.88}
  {'units': 2068, 'category': 'Carney-Santos', 'month': '2026-08', 'revenue': 505719.07, 'est_profit': 129412.48}
  {'units': 1915, 'category': 'Rodriguez, Figueroa and Sanchez', 'month': '2026-08', 'revenue': 498684.96, 'est_profit': 87307.6}
  {'units': 2049, 'category': 'Burns-Rodriguez', 'month': '2026-08', 'revenue': 488537.97, 'est_profit': 88694.19}

== 3a. Users by purchase frequency (bucket lower bound = orders) ===
  {'orders_bucket': 1, 'users': 1179, 'avg_spent': 1833.38, 'total_revenue': 2161560.82}
  {'orders_bucket': 3, 'users': 4907, 'avg_spent': 4353.7, 'total_revenue': 21363585.65}
  {'orders_bucket': 6, 'users': 3550, 'avg_spent': 7351.23, 'total_revenue': 26096881.28}
  {'orders_bucket': 10, 'users': 313, 'avg_spent': 11449.47, 'total_revenue': 3583682.6}

== 3b. Users by total spend (bucket lower bound = $) ===
  {'spend_bucket': 0, 'users': 101, 'avg_orders': 1.25}
  {'spend_bucket': 500, 'users': 499, 'avg_orders': 1.99}
  {'spend_bucket': 1500, 'users': 1558, 'avg_orders': 2.95}
  {'spend_bucket': 3000, 'users': 4112, 'avg_orders': 4.59}
  {'spend_bucket': 6000, 'users': 3679, 'avg_orders': 6.91}

== 3c. Top customers ===
  {'_id': 'user_001797', 'orders': 13, 'spent': 22742.16, 'last_order': datetime.datetime(2026, 11, 6, 21, 27, 27)}
  {'_id': 'user_000347', 'orders': 14, 'spent': 20601.76, 'last_order': datetime.datetime(2026, 11, 8, 2, 8, 1)}
  {'_id': 'user_006035', 'orders': 15, 'spent': 20080.4, 'last_order': datetime.datetime(2026, 11, 7, 20, 13, 26)}
  {'_id': 'user_001241', 'orders': 11, 'spent': 18490.51, 'last_order': datetime.datetime(2026, 11, 5, 2, 27, 9)}
  {'_id': 'user_008354', 'orders': 10, 'spent': 18308.51, 'last_order': datetime.datetime(2026, 11, 4, 9, 36, 23)}

== 4. Session funnel by referrer ===
  {'_id': 'direct', 'sessions': 9984, 'viewed_product': 9858, 'reached_cart': 6803, 'reached_checkout': 4440, 'converted': 2088, 'conversion_rate': 0.21}
  {'_id': 'search_engine', 'sessions': 10107, 'viewed_product': 9984, 'reached_cart': 6934, 'reached_checkout': 4507, 'converted': 2101, 'conversion_rate': 0.21}
  {'_id': 'social', 'sessions': 10006, 'viewed_product': 9871, 'reached_cart': 6778, 'reached_checkout': 4409, 'converted': 2054, 'conversion_rate': 0.21}
  {'_id': 'affiliate', 'sessions': 9967, 'viewed_product': 9837, 'reached_cart': 6736, 'reached_checkout': 4357, 'converted': 2022, 'conversion_rate': 0.2}
  {'_id': 'email', 'sessions': 9936, 'viewed_product': 9818, 'reached_cart': 6774, 'reached_checkout': 4353, 'converted': 1999, 'conversion_rate': 0.2}
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics>
```

Comment: Top sellers (84 units for the leader); revenue and estimated profit per category and month; frequency and spend segments (313 users with 10+ orders average $11,449); top customers; journey funnel by referrer (20-21% flagged converted for every referrer).

S7 - mongosh: counts and pipeline 1

Command: `Get-Content mongodb\evidence.js | docker exec -i mongo mongosh ecommerce --quiet`

```
=== 1. MongoDB: document count + pipeline 1 (top-selling products) ===
ecommerce>
ecommerce>
ecommerce> === Document counts ===

ecommerce> transactions: 50000

ecommerce> users:        10000

ecommerce> products:     5000

ecommerce>
=== Pipeline 1: top-selling products ===

ecommerce> | | | | | | | | | | | [
  {
    product_name: 'Right-Sized Bifurcated Help-Desk',
    units_sold: 84,
    orders: 35,
    product_id: 'prod_03248',
    revenue: 31485.72,
    current_stock: 816
  },
  {
    product_name: 'Cloned Discrete Array',
    units_sold: 84,
    orders: 36,
    product_id: 'prod_04028',
    revenue: 12798.24,
    current_stock: 241
  },
  {
    product_name: 'Centralized Executive Customer Loyalty',
    units_sold: 82,
    orders: 35,
    product_id: 'prod_01324',
    revenue: 15967.04,
    current_stock: 589
  },
  {
    product_name: 'Optional Solution-Oriented Concept',
    units_sold: 81,
    orders: 36,
    product_id: 'prod_00810',
    revenue: 30923.37,
    current_stock: 918
  },
  {
    product_name: 'Inverse Stable Paradigm',
    units_sold: 81,
    orders: 38,
    product_id: 'prod_04620',
    revenue: 22634.64,
    current_stock: 724
  },
  {
    product_name: 'Universal Methodical Synergy',
    units_sold: 81,
    orders: 37,
    product_id: 'prod_03004',
    revenue: 19151.64,
    current_stock: 877
  },
  {
    product_name: 'Networked Next Generation Capability',
    units_sold: 81,
    orders: 40,
    product_id: 'prod_00366',
    revenue: 12248.01,
    current_stock: 659
  },
  {
{
  product_name: 'Phased Well-Modulated Encoding',
  units_sold: 80,
  orders: 35,
  product_id: 'prod_04258',
  revenue: 516,
  current_stock: 478
},
{
  product_name: 'Balanced Foreground Internet Solution',
  units_sold: 79,
  orders: 37,
  product_id: 'prod_04961',
  revenue: 29087.8,
  current_stock: 48
},
{
  product_name: 'Re-Contextualized Scalable Access',
  units_sold: 79,
  orders: 37,
  product_id: 'prod_03342',
  revenue: 28309.65,
  current_stock: 288
}
```

Comment: Run directly in the MongoDB shell: 50,000 transactions, 10,000 users, 5,000 products; the top-selling pipeline with $lookup to current stock returns the same units, orders and revenue as the Python run (S6).

C.4 Part 1 - HBase

S8 - HBase shell: list, describe and a user prefix scan

Command: `Get-Content hbase\evidence.hbase | docker exec -i hbase hbase shell -n`



> **Comment:** Both tables listed; describe shows VERSIONS 1, ROW bloom filters and the 180-day TTL on pv. The prefix scan of user_000042| returns the user's newest sessions first without sorting - the reversed timestamp in the row key at work.

C.5 Part 2 - Spark processing

S9 - Cleaning job and data-quality report

Command: `docker compose exec -T app python spark/01_clean_normalize.py`

```
=== 3. Spark cleaning job (data-quality counts) ===
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/11/10 12:42:20 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
categories
   [categories] subcategory_rows: 109
users
   [users] raw_rows: 10,000
   [users] duplicates_removed: 0
   [users] clean_rows: 10,000
products
   [products] duplicates_removed: 0
   [products] missing_subcategory_filled: 0
   [products] negative_price_or_stock: 0
sessions (all sessions_*.json files)
   [sessions] raw_rows: 200,000
   [sessions] duplicates_removed: 0
   [sessions] invalid_rows_dropped: 0
   [sessions] orphan_user_id: 0
   [sessions] status_repaired_abandoned_to_browsed: 0
transactions
   [transactions] raw_rows: 50,000
   [transactions] duplicates_removed: 0
   [transactions] total_mismatch_flagged: 0
   [transactions] items_sum_mismatch_flagged: 0
   [transactions] orphan_user_id: 0
   [transaction_items] orphan_product_id: 0
   [sessions] converted_without_transaction: 15,552
   [users] last_active_before_registration: 0

Clean tables written to /app/data/clean in 1190s

--- data quality report ---
table,check,count
categories,subcategory_rows,109
users,raw_rows,10000
users,duplicates_removed,0
users,clean_rows,10000
products,duplicates_removed,0
products,missing_subcategory_filled,0
products,negative_price_or_stock,0
sessions,raw_rows,200000
sessions,duplicates_removed,0
sessions,invalid_rows_dropped,0
sessions,orphan_user_id,0
sessions,status_repaired_abandoned_to_browsed,0
transactions,raw_rows,50000
transactions,duplicates_removed,0
transactions,total_mismatch_flagged,0
transactions,items_sum_mismatch_flagged,0
transactions,orphan_user_id,0
transaction_items,orphan_product_id,0
sessions,converted_without_transaction,15552
users,last_active_before_registration,0
```

> **Comment:** All 200,000 sessions, 50,000 transactions and 10,000 users validated: 0 duplicates, 0 invalid rows, 0 orphans, 0 total mismatches; 15,552 converted sessions without an order are flagged, not deleted. Runtime 1,190 s.

S10 - Recommendation indicators

Command: `docker compose exec -T app python spark/02_recommendations.py`

```
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/11/10 15:44:37 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

== Bought together (product): 50,000 baskets, 304 pairs with count >= 2
+----------+----------+----------+-------+-------+-------+--------------------+------------------+----+
|y         |x         |pair_count|x_count|y_count|support|confidence          |lift              |rank|
+----------+----------+----------+-------+-------+-------+--------------------+------------------+----+
|prod_00256|prod_00377|2         |10     |23     |4.0E-5 |0.2                 |434.78260869565522|1   |
|prod_00377|prod_00256|2         |23     |10     |4.0E-5 |0.08695652173913043 |434.78260869565213|1   |
|prod_03078|prod_01514|2         |11     |23     |4.0E-5 |0.18181818181818182 |395.25691699604744|1   |
|prod_01514|prod_03078|2         |23     |11     |4.0E-5 |0.08695652173913043 |395.2569169960474 |1   |
|prod_03071|prod_04513|2         |20     |15     |4.0E-5 |0.1                 |333.33333333333337|1   |
|prod_04513|prod_03071|2         |15     |20     |4.0E-5 |0.13333333333333333 |333.3333333333333 |1   |
|prod_04016|prod_03720|2         |21     |15     |4.0E-5 |0.09523809523809523 |317.46031746031747|1   |
|prod_03720|prod_04016|2         |15     |21     |4.0E-5 |0.13333333333333333 |317.46031746031747|1   |
|prod_04871|prod_04232|2         |32     |10     |4.0E-5 |0.0625              |312.5             |1   |
|prod_04232|prod_04871|2         |10     |32     |4.0E-5 |0.2                 |312.5             |1   |
+----------+----------+----------+-------+-------+-------+--------------------+------------------+----+
only showing top 10 rows


== Viewed together (product): 197,509 baskets, 59,127 pairs with count >= 2
+----------+----------+----------+-------+-------+--------------------+--------------------+------------------+----+
|y         |x         |pair_count|x_count|y_count|support             |confidence          |lift              |rank|
+----------+----------+----------+-------+-------+--------------------+--------------------+------------------+----+
|prod_01165|prod_01434|5         |149    |150    |2.5315302087499818E-5|0.03355704697986577 |44.1854586129753951|1   |
|prod_01434|prod_01165|5         |150    |149    |2.5315302087499818E-5|0.03333333333333333 |44.185458612975391 |1   |
|prod_00741|prod_01248|4         |150    |109    |2.0252241669999847E-5|0.02666666666666667 |48.32024464831805  |1   |
|prod_01248|prod_00741|4         |109    |150    |2.0252241669999847E-5|0.03669724770642202 |48.32024464831804 5|1   |
|prod_04997|prod_02035|4         |117    |156    |2.0252241669999847E-5|0.03418803418803419 |43.28490028490029  |1   |
|prod_02035|prod_04997|4         |156    |117    |2.0252241669999847E-5|0.02564102564102564 |43.28490028490028  |1   |
|prod_04543|prod_01294|4         |132    |146    |2.0252241669999847E-5|0.030303030303030304|40.993980904939381 |1   |
|prod_01294|prod_04543|4         |146    |132    |2.0252241669999847E-5|0.02739726027397726 |40.993980904939804 |1   |
|prod_04616|prod_04510|4         |148    |133    |2.0252241669999847E-5|0.02702702702702703 |40.135947978053245 |1   |
|prod_04510|prod_04616|4         |133    |148    |2.0252241669999847E-5|0.03007518796992481 |40.135947978053244 |1   |
+----------+----------+----------+-------+-------+--------------------+--------------------+------------------+----+
only showing top 10 rows


== Bought together (category): 50,000 baskets, 300 pairs with count >= 1
+--------------------------+--------------------------+----------+-------+-------+-------+--------------------+------------------+----+
|y                         |x                         |pair_count|x_count|y_count|support|confidence          |lift              |rank|
+--------------------------+--------------------------+----------+-------+-------+-------+--------------------+------------------+----+
|Skinner-Romero            |Mitchell-Kim              |335       |4803   |4351   |0.0067 |0.06974807412034145 |0.8015177444304924|2   |
|Mitchell-Kim              |Skinner-Romero            |335       |4351   |4803   |0.0067 |0.07699379452999311 |0.8015177444304924|1   |
|Owens-Daniel              |Mitchell-Kim              |332       |4803   |4272   |0.00664|0.06912346450135332 |0.8090293129840042|1   |
|Mitchell-Kim              |Owens-Daniel              |332       |4272   |4803   |0.00664|0.07771535580524344 |0.8090293129840042|1   |
|Rodriguez-Graham          |Burns-Rodriguez           |332       |4629   |4464   |0.00664|0.07172175415856556 |0.8033350600197756|3   |
|Burns-Rodriguez           |Rodriguez-Graham          |332       |4464   |4629   |0.00664|0.07437275985663082 |0.8033350600197756|1   |
|Rodriguez-Graham          |Cameron-Parsons           |325       |4591   |4464   |0.0065 |0.07079067741232847 |0.7929063330233924|1   |
|Cameron-Parsons           |Rodriguez-Graham          |325       |4464   |4591   |0.0065 |0.07280465949820789 |0.7929063330233923|2   |
|Moore, Becker and Carlson |Anderson, Walls and Duncan|324       |4429   |4536   |0.00648|0.07315421088281779 |0.8063735767506371 |2   |
|Anderson, Walls and Duncan|Moore, Becker and Carlson |324       |4536   |4429   |0.00648|0.07142857142857142 |0.8063735767506369 |3   |
+--------------------------+--------------------------+----------+-------+-------+-------+--------------------+------------------+----+
only showing top 10 rows

+------+---+-------+----------+--------+
| rules|  n|avg_lift|median_lift|max_lift|
+------+---+-------+----------+--------+
|bought|608|  166.97|     155.28|  434.78|
+------+---+-------+----------+--------+

+------+-----+-------+----------+--------+
| rules|    n|avg_lift|median_lift|max_lift|
+------+-----+-------+----------+--------+
|viewed|23430|  22.588|    20.743|  196.04|
+------+-----+-------+----------+--------+

+--------+---+-------+----------+--------+
|   rules|  n|avg_lift|median_lift|max_lift|
+--------+---+-------+----------+--------+
|category|125|  0.785|     0.781|    0.83|
+--------+---+-------+----------+--------+
```

Comment: 304 product pairs occur twice or more in 50,000 baskets (lifts up to 435 from tiny counts); 59,127 viewed-together pairs; category lift ~0.79. Conclusion: no real affinity in random synthetic data - the method needs support thresholds on real data.

S11 - Cohort analysis

Command: `docker compose exec -T app python spark/03_cohort_analysis.py`

```
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics> docker compose exec -T app python spark/03_cohort_analysis.py
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/11/10 15:51:18 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
+-----------------+---------+---------------+-------------+------+----------+------------+-----------+----------------+-------+----------------------+
|registration_month|txn_month|months_since_reg|active_buyers|orders|revenue   |cohort_users|retention_pct|revenue_per_member|aov    |cum_revenue_per_member|
+-----------------+---------+---------------+-------------+------+----------+------------+-----------+----------------+-------+----------------------+
|2026-02          |2026-08  |6              |595          |958   |1027724.22|891         |66.8       |1153.45         |1072.78|1153.45               |
|2026-02          |2026-09  |7              |726          |1464  |1597389.3 |891         |81.5       |1792.81         |1091.11|2946.26               |
|2026-02          |2026-10  |8              |733          |1548  |1615599.09|891         |82.3       |1813.24         |1043.67|4759.5                |
|2026-02          |2026-11  |9              |336          |418   |441666.15 |891         |37.7       |495.7           |1056.62|5255.2                |
|2026-03          |2026-08  |5              |1175         |1904  |1987402.95|1775        |66.2       |1119.66         |1043.8 |1119.66               |
|2026-03          |2026-09  |6              |1433         |2900  |3066710.71|1775        |80.7       |1727.72         |1057.49|2847.38               |
|2026-03          |2026-10  |7              |1487         |3123  |3322095.32|1775        |83.8       |1871.6          |1063.75|4718.98               |
|2026-03          |2026-11  |8              |702          |884   |948039.33 |1775        |39.5       |534.11          |1072.44|5253.09               |
|2026-04          |2026-08  |4              |1112         |1876  |1989037.54|1662        |66.9       |1196.77         |1060.25|1196.77               |
|2026-04          |2026-09  |5              |1326         |2794  |2978453.37|1662        |79.8       |1792.89         |1066.02|2988.86               |
|2026-04          |2026-10  |6              |1380         |2904  |3068965.53|1662        |83.0       |1846.55         |1056.81|4835.41               |
|2026-04          |2026-11  |7              |682          |836   |888508.77 |1662        |41.0       |534.6           |1062.81|5370.01               |
|2026-05          |2026-08  |3              |1172         |1939  |2096537.47|1735        |67.6       |1208.38         |1081.25|1208.38               |
|2026-05          |2026-09  |4              |1407         |2903  |3106378.08|1735        |81.1       |1790.42         |1070.06|2998.8                |
|2026-05          |2026-10  |5              |1426         |2984  |3096650.57|1735        |82.2       |1784.81         |1037.75|4783.61               |
|2026-05          |2026-11  |6              |692          |909   |988871.43 |1735        |39.9       |569.95          |1087.87|5353.56               |
|2026-06          |2026-08  |2              |1121         |1858  |1976459.68|1635        |68.6       |1208.84         |1063.76|1208.84               |
|2026-06          |2026-09  |3              |1332         |2724  |2874887.61|1635        |81.5       |1758.34         |1055.39|2967.18               |
|2026-06          |2026-10  |4              |1350         |2805  |2997012.35|1635        |82.6       |1833.04         |1068.45|4800.22               |
|2026-06          |2026-11  |5              |666          |846   |924308.86 |1635        |40.7       |565.33          |1092.56|5365.55               |
|2026-07          |2026-08  |1              |1121         |1853  |1979964.34|1671        |67.1       |1184.9          |1068.52|1184.9                |
|2026-07          |2026-09  |2              |1358         |2837  |3064376.41|1671        |81.3       |1833.86         |1080.15|3018.76               |
|2026-07          |2026-10  |3              |1361         |2827  |2976952.3 |1671        |81.4       |1781.54         |1053.04|4800.3                |
|2026-07          |2026-11  |4              |618          |786   |851649.68 |1671        |37.0       |509.66          |1083.52|5309.96               |
|2026-08          |2026-08  |0              |416          |696   |723605.13 |631         |65.9       |1146.76         |1039.66|1146.76               |
|2026-08          |2026-09  |1              |510          |1061  |1141852.51|631         |80.8       |1809.59         |1076.2 |2956.35               |
|2026-08          |2026-10  |2              |507          |1041  |1129454.53|631         |80.3       |1789.94         |1084.97|4746.29               |
|2026-08          |2026-11  |3              |258          |322   |345157.12 |631         |40.9       |547.0           |1071.92|5293.29               |
+-----------------+---------+---------------+-------------+------+----------+------------+-----------+----------------+-------+----------------------+

Retention % (rows = registration cohort, columns = purchase month)
+-----------------+------------+------+------+------+------+
|registration_month|cohort_users|2026-08|2026-09|2026-10|2026-11|
+-----------------+------------+------+------+------+------+
|2026-02          |891         |66.8  |81.5  |82.3  |37.7  |
|2026-03          |1775        |66.2  |80.7  |83.8  |39.5  |
|2026-04          |1662        |66.9  |79.8  |83.0  |41.0  |
|2026-05          |1735        |67.6  |81.1  |82.2  |39.9  |
|2026-06          |1635        |68.6  |81.5  |82.6  |40.7  |
|2026-07          |1671        |67.1  |81.3  |81.4  |37.0  |
|2026-08          |631         |65.9  |80.8  |80.3  |40.9  |
+-----------------+------------+------+------+------+------+

+-----------------+------------+-----------------------+-------+
|registration_month|cohort_users|revenue_per_member_90d|avg_aov|
+-----------------+------------+-----------------------+-------+
|          2026-02|         891|                5255.2|1066.05|
|          2026-03|        1775|                5253.1|1059.37|
|          2026-04|        1662|               5370.02|1061.47|
|          2026-05|        1735|               5353.57|1069.23|
|          2026-06|        1635|               5365.55|1070.04|
|          2026-07|        1671|               5309.96|1071.31|
|          2026-08|         631|                5293.3|1068.19|
+-----------------+------------+-----------------------+-------+
```

Comment: Seven registration cohorts; about 80-84% of each buys in each full month (Aug and Nov are partial); 90-day revenue per member is flat at $5,253-$5,370.

S12 - Spark SQL analytics

Command: `docker compose exec -T app python spark/04_spark_sql_analytics.py`

```
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics> docker compose exec -T app python spark/04_spark_sql_analytics.py
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/11/10 15:54:17 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

==================== q1_top3_products_per_category
+--------------------------+---+----------+----------------------------------------------+-----+--------+----------+--------------+
|category_name             |rnk|product_id|product_name                                  |units|revenue |est_profit|pct_of_category|
+--------------------------+---+----------+----------------------------------------------+-----+--------+----------+--------------+
|Anderson, Walls and Duncan|1  |prod_01092|De-Engineered Object-Oriented Parallelism     |70   |36545.28|13521.79  |1.69          |
|Anderson, Walls and Duncan|2  |prod_02000|Public-Key Full-Range Info-Mediaries          |69   |32720.46|7198.48   |1.51          |
|Anderson, Walls and Duncan|3  |prod_01571|Automated Global Approach                     |57   |30686.32|11354.02  |1.42          |
|Baldwin Ltd               |1  |prod_03014|Up-Sized Zero Administration Parallelism      |69   |27356.22|6291.85   |1.29          |
|Baldwin Ltd               |2  |prod_02144|Persistent Regional Data-Warehouse            |59   |27068.15|2977.45   |1.27          |
|Baldwin Ltd               |3  |prod_04922|Persevering Web-Enabled Initiative            |62   |27052.07|10009.27  |1.27          |
|Brewer, Bryant and Hickman|1  |prod_01290|Up-Sized Fresh-Thinking Utilization           |65   |33574.11|3357.39   |1.64          |
|Brewer, Bryant and Hickman|2  |prod_03941|Centralized Multimedia Toolset                |56   |30592.58|9177.77   |1.49          |
|Brewer, Bryant and Hickman|3  |prod_00980|Virtual Dynamic Focus Group                   |57   |28727.91|7469.21   |1.4           |
|Burns-Rodriguez           |1  |prod_00810|Optional Solution-Oriented Concept            |81   |30446.16|4871.42   |1.34          |
|Burns-Rodriguez           |2  |prod_04587|Sharable Scalable Interface                   |55   |29203.89|4672.65   |1.28          |
|Burns-Rodriguez           |3  |prod_02786|Customizable Directional Budgetary Management|59   |29143.0 |6119.98   |1.28          |
|Bush Inc                  |1  |prod_02588|Up-Sized Global Intranet                      |60   |31570.03|10733.74  |1.67          |
|Bush Inc                  |2  |prod_01243|Diverse Modular Array                         |76   |31443.69|10690.86  |1.66          |
|Bush Inc                  |3  |prod_02251|ProFound Bi-Directional Graphic Interface     |75   |27889.93|9482.66   |1.47          |
+--------------------------+---+----------+----------------------------------------------+-----+--------+----------+--------------+
only showing top 15 rows


==================== q2_funnel_by_device_referrer
+-----------+-----------+--------+-----------------+-----------------+-------------+-----------------+-----------------------+----------+
|device_type|referrer   |sessions|pct_product_view|pct_add_to_cart|pct_checkout|pct_converted_flag|pct_with_transaction|avg_minutes|
+-----------+-----------+--------+-----------------+-----------------+-------------+-----------------+-----------------------+----------+
|desktop    |affiliate  |13307   |98.7             |65.1             |43.9         |20.67            |12.82                  |30.3       |
|desktop    |direct     |13272   |98.6             |65.1             |43.2         |20.52            |12.69                  |30.2       |
|desktop    |email      |13212   |98.9             |64.7             |43.5         |20.13            |12.22                  |30.4       |
|desktop    |search_engine|13303 |98.6             |65.4             |43.3         |20.47            |12.81                  |30.2       |
|desktop    |social     |13387   |98.7             |64.7             |43.9         |20.23            |12.49                  |30.4       |
|desktop    |NULL       |66481   |98.7             |65.0             |43.6         |20.4             |12.61                  |30.3       |
|mobile     |affiliate  |13373   |98.8             |65.2             |43.4         |20.1             |12.78                  |30.3       |
|mobile     |direct     |13516   |98.7             |64.9             |44.5         |20.53            |12.8                   |30.2       |
|mobile     |email      |13265   |98.8             |65.6             |43.5         |20.38            |12.33                  |30.5       |
|mobile     |search_engine|13375 |98.8             |65.1             |43.0         |20.42            |12.8                   |30.2       |
|mobile     |social     |13431   |98.8             |65.6             |43.4         |20.21            |12.38                  |29.9       |
|mobile     |NULL       |66960   |98.8             |65.3             |43.8         |20.33            |12.62                  |30.2       |
|tablet     |affiliate  |13210   |98.8             |65.7             |43.7         |20.41            |12.15                  |30.1       |
|tablet     |direct     |13302   |98.8             |64.7             |43.7         |20.22            |12.52                  |30.0       |
|tablet     |email      |13642   |98.8             |64.8             |44.5         |21.03            |13.25                  |30.3       |
|tablet     |search_engine|13089 |98.7             |64.3             |44.3         |20.4             |12.76                  |30.3       |
|tablet     |social     |13316   |98.7             |64.9             |43.6         |19.89            |12.16                  |30.3       |
|tablet     |NULL       |66559   |98.8             |64.9             |44.0         |20.39            |12.57                  |30.2       |
|NULL       |NULL       |200000  |98.8             |65.1             |43.8         |20.37            |12.6                   |30.3       |
+-----------+-----------+--------+-----------------+-----------------+-------------+-----------------+-----------------------+----------+


==================== q3_daily_revenue_rolling
+----------+------+---------+---------+--------------+--------------+-------+
|txn_date  |orders|revenue  |discounts|revenue_7d_avg|wow_change_pct|weekday|
+----------+------+---------+---------+--------------+--------------+-------+
|2026-08-12|511   |552184.81|14470.05 |552184.81     |NULL          |Wed    |
|2026-08-13|502   |517508.13|11606.13 |534846.47     |NULL          |Thu    |
|2026-08-14|533   |595342.63|17799.4  |555011.86     |NULL          |Fri    |
|2026-08-15|555   |612408.57|17246.78 |569361.03     |NULL          |Sat    |
|2026-08-16|524   |529027.29|12941.81 |561294.29     |NULL          |Sun    |
|2026-08-17|554   |582127.03|15406.06 |564766.41     |NULL          |Mon    |
|2026-08-18|595   |596030.62|15092.0  |569232.73     |NULL          |Tue    |
|2026-08-19|552   |587297.54|15902.7  |574248.83     |6.4           |Wed    |
|2026-08-20|553   |589466.92|14102.61 |584528.66     |13.9          |Thu    |
|2026-08-21|570   |614339.78|17091.53 |587242.54     |3.2           |Fri    |
+----------+------+---------+---------+--------------+--------------+-------+
only showing top 10 rows

==================== q4_price_change_impact
+--------------+--------+-----------------------+----------------------+
|price_move    |products|units_per_day_before|units_per_day_after|
+--------------+--------+-----------------------+----------------------+
|1. cut >10%   |573     |0.509                  |0.49                  |
|2. cut 0-10%  |525     |0.496                  |0.481                 |
|3. raise 0-10%|575     |0.509                  |0.484                 |
|4. raise >10% |602     |0.514                  |0.493                 |
+--------------+--------+-----------------------+----------------------+


==================== q5_viewed_vs_sold
+----------+--------------------------------------------+--------------------------------+-------------+-------------+---------+-----+-----+-----+-----+----------+-----------+-----------+
|product_id|product_name                                |category_name                   |current_price|current_stock|is_active|views|cart_adds|orders|units_sold|revenue   |view_to_cart|view_rank|sales_rank|
+----------+--------------------------------------------+--------------------------------+-------------+-------------+---------+-----+---------+------+----------+----------+------------+---------+----------+
|prod_03029|Cross-Platform Demand-Driven Neural-Net      |Burns-Rodriguez                 |253.52       |133          |true     |172  |50       |16    |31        |7516.86   |0.291       |1        |4348      |
|prod_04028|Cloned Discrete Array                       |Patterson, Smith and Jones      |152.36       |241          |true     |170  |58       |36    |84        |12432.59  |0.341       |2        |1         |
|prod_01064|Fundamental Upward-Trending Product         |Carney-Santos                   |131.1        |233          |true     |170  |56       |30    |62        |7833.23   |0.329       |2        |325       |
|prod_02511|Cloned Mission-Critical Instruction Set     |Wright-Leblanc                  |272.06       |189          |true     |169  |60       |26    |50        |13385.35  |0.355       |4        |1578      |
|prod_01268|Upgradable Asynchronous Framework           |Skinner-Romero                  |182.07       |548          |true     |168  |59       |35    |75        |13409.46  |0.351       |5        |25        |
|prod_03016|Synergistic Solution-Oriented Knowledgebase |Mitchell-Kim                    |351.55       |784          |true     |168  |55       |29    |58        |19968.05  |0.327       |5        |581       |
|prod_03677|Diverse Holistic Support                    |Hancock Inc                     |75.69        |511          |true     |167  |51       |31    |57        |4159.16   |0.305       |7        |664       |
|prod_04525|Proactive Reciprocal Monitoring             |Mitchell-Kim                    |331.23       |747          |true     |167  |51       |23    |39        |12454.25  |0.305       |7        |3506      |
|prod_02890|Organic Well-Modulated Hub                  |Mitchell-Kim                    |459.08       |567          |true     |167  |59       |22    |36        |16182.58  |0.353       |7        |3899      |
|prod_01421|Realigned National Adapter                  |King-Odonnell                   |368.98       |347          |true     |167  |54       |17    |33        |11807.36  |0.323       |7        |4183      |
+----------+--------------------------------------------+--------------------------------+-------------+-------------+---------+-----+---------+------+----------+----------+------------+---------+----------+
only showing top 10 rows

################## Part B - querying 'MongoDB' and 'HBase' shaped data

==================== qB1_engagement_vs_value_mongo_x_hbase
+---------------+-----+----------+---------+---------------+
|engagement_band|users|avg_orders|avg_spent|avg_session_min|
+---------------+-----+----------+---------+---------------+
|1. <15         |1057 |4.15      |4730.67  |30.3           |
|2. 15-19       |3674 |4.64      |5052.49  |30.1           |
|3. 20-24       |3706 |5.21      |5463.81  |30.3           |
|4. 25+ sessions|1563 |5.93      |6010.02  |30.3           |
+---------------+-----+----------+---------+---------------+


==================== qB2_category_revenue_from_embedded_items
+-----------+-------------+------+
|category_id|gross_revenue|orders|
+-----------+-------------+------+
|cat_021    |852720.54    |1620  |
|cat_007    |818234.3     |1515  |
|cat_024    |813597.09    |1537  |
|cat_002    |813282.89    |1409  |
|cat_005    |808483.64    |1517  |
|cat_010    |798986.81    |1534  |
|cat_000    |785467.62    |1455  |
|cat_008    |759978.58    |1484  |
|cat_009    |756323.4     |1416  |
|cat_004    |740362.87    |1509  |
+-----------+-------------+------+


==================== qB3_product_view_trend_hbase
+----------+-------------------+-----+
|product_id|week               |views|
+----------+-------------------+-----+
|prod_01064|2026-08-10 00:00:00|7    |
|prod_01064|2026-08-17 00:00:00|13   |
|prod_01064|2026-08-24 00:00:00|10   |
|prod_01064|2026-08-31 00:00:00|10   |
|prod_01064|2026-09-07 00:00:00|13   |
|prod_01064|2026-09-14 00:00:00|15   |
|prod_01064|2026-09-21 00:00:00|11   |
|prod_01064|2026-09-28 00:00:00|10   |
|prod_01064|2026-10-05 00:00:00|21   |
|prod_01064|2026-10-12 00:00:00|12   |
+----------+-------------------+-----+
only showing top 10 rows
```

Comment: Part A: top-3 per category, ROLLUP funnel (12.6% of 200,000 sessions have an order), 7-day rolling revenue, price-change impact, viewed-vs-sold ranks. Part B: MongoDB-shaped users joined with HBase-shaped sessions - spend rises from $4,731 (<15 sessions) to $6,010 (25+).

C.6 Part 3 - Integrated analytics

S13 - Customer Lifetime Value (MongoDB + HBase via Spark)

Command: `docker compose exec -T app python integration/clv_analysis.py --mongo mongo --hbase hbase`

```
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics> docker compose exec -T app python integration/clv_analysis.py --mongo mongo --hbase hbase
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/11/10 18:00:09 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
   user_sessions: 20/100 key ranges scanned
   user_sessions: 40/100 key ranges scanned
   user_sessions: 60/100 key ranges scanned
   user_sessions: 80/100 key ranges scanned
   user_sessions: 100/100 key ranges scanned
   scanned 200,000 session rows from HBase (hbase)
Observation window: 2026-08-12 .. 2026-11-10 (90 days)
Monthly buyer retention r = 0.830 -> expected lifetime = 5.9 months
+---------------------+-----+-------+---------+---------------+----------+----------------+------------+-------------------------+----------------+
|segment              |users|avg_clv|total_clv|avg_revenue_90d|avg_orders|avg_recency_days|avg_sessions|avg_session_purchase_rate|share_of_clv_pct|
+---------------------+-----+-------+---------+---------------+----------+----------------+------------+-------------------------+----------------+
|Champions            |1729 |4328.77|7484438.0|8586.25        |7.56      |3.9             |21.1        |0.171                    |27.9            |
|At risk - high value|1084 |3910.18|4238639.0|7776.39        |6.23      |26.7            |20.2        |0.143                    |15.8            |
|Loyal                |1260 |3394.85|4277508.0|6723.56        |6.93      |9.4             |21.1        |0.179                    |16.0            |
|Needs attention      |1939 |2281.56|4423935.0|4531.25        |4.24      |8.9             |19.6        |0.109                    |16.5            |
|New / promising      |1041 |1662.93|1731107.0|3300.24        |3.25      |4.3             |19.1        |0.091                    |6.5             |
|Hibernating          |2896 |1607.66|4655778.0|3189.62        |3.39      |35.0            |19.4        |0.095                    |17.4            |
|No purchase          |51   |0.0    |0.0      |0.0            |0.0       |90.0            |17.5        |0.0                      |0.0             |
+---------------------+-----+-------+---------+---------------+----------+----------------+------------+-------------------------+----------------+

+---------+-----+-------+------+--------+------------+---------------+--------+--------+---------------------+-------+----------+
|clv_tier|users|avg_clv|min_clv|max_clv |avg_sessions|avg_session_min|avg_pages|cart_rate|session_purchase_rate|avg_aov|avg_orders|
+---------+-----+-------+------+--------+------------+---------------+--------+--------+---------------------+-------+----------+
|Bronze  |2488 |1025.31|4.74  |1615.85 |19.3        |30.2           |9.9     |0.634   |0.09                 |829.34 |2.85      |
|Silver  |2487 |2075.54|1615.91|2528.81 |19.8        |30.2           |10.0    |0.647   |0.119                |1031.84|4.39      |
|Gold    |2487 |3013.9 |2529.25|3585.25 |20.1        |30.2           |10.0    |0.655   |0.135                |1138.0 |5.53      |
|Platinum|2487 |4665.47|3585.58|11078.62|20.9        |30.4           |10.1    |0.666   |0.162                |1264.65|7.33      |
+---------+-----+-------+------+--------+------------+---------------+--------+--------+---------------------+-------+----------+

+------------------+-----------------+------------------+
|corr_sessions_clv|corr_duration_clv|corr_cart_rate_clv|
+------------------+-----------------+------------------+
|             0.133|            0.017|             0.111|
+------------------+-----------------+------------------+
```

Comment: Spark read orders from MongoDB and all 200,000 sessions from HBase in 100 row-key ranges, estimated retention r = 0.830 (5.9-month lifetime) and produced the segment, tier and correlation tables of §5.1.

S14 - Funnel conversion (HBase journeys + MongoDB purchases)

Command: `docker compose exec -T app python integration/funnel_analysis.py --mongo mongo --hbase hbase`

```
(.venv) PS C:\Users\Hp\OneDrive - Zaria Group\Documents\Masters\Big Data Analytics\final Exam\E-commerce-analytics> docker compose exec -T app python integration/funnel_analysis.py --mongo mongo --hbase hbase
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/11/10 18:11:56 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
   user_sessions: 20/100 key ranges scanned
   user_sessions: 40/100 key ranges scanned
   user_sessions: 60/100 key ranges scanned
   user_sessions: 80/100 key ranges scanned
   user_sessions: 100/100 key ranges scanned
   scanned 200,000 session rows from HBase (hbase)
   product_metrics: 20/50 key ranges scanned
   product_metrics: 40/50 key ranges scanned
   scanned 342,423 product-day rows from HBase (hbase)
+---------------+--------+---------------+------------------+-------+
|stage          |sessions|pct_of_sessions|step_conversion_pct|dropped|
+---------------+--------+---------------+------------------+-------+
|1. Session     |200000  |100.0          |100.0             |0      |
|2. Product view|197509  |98.75          |98.75             |2491   |
|3. Add to cart |130126  |65.06          |65.88             |67383  |
|4. Checkout    |58215   |29.11          |44.74             |71911  |
|5. Purchase    |25197   |12.6           |43.28             |33018  |
+---------------+--------+---------------+------------------+-------+

Sessions flagged 'converted' by the generator: 40,749 vs sessions with a real transaction: 25,197
Average web order value $771.02; checkout abandoners 33,018 => ~$25,457,514 unrealised revenue in the window
+-----------+--------+--------+------------+-----------+------------------+
|device_type|sessions|cart_pct|checkout_pct|purchase_pct|revenue_per_session|
+-----------+--------+--------+------------+-----------+------------------+
|     mobile|   66960|   65.31|       29.19|      12.62|              97.5|
|    desktop|   66481|    65.0|       29.04|      12.61|             96.03|
|     tablet|   66559|   64.88|       29.09|      12.57|             97.88|
+-----------+--------+--------+------------+-----------+------------------+

+--------------+--------+--------+------------+------------+------------------+
|     referrer|sessions|cart_pct|checkout_pct|purchase_pct|revenue_per_session|
+--------------+--------+--------+------------+------------+------------------+
|search_engine|   39767|   64.93|       29.18|       12.79|             97.45|
|       direct|   40090|   64.94|        29.1|       12.67|             96.87|
|        email|   40119|   65.04|       29.16|       12.61|             97.48|
|    affiliate|   39890|   65.32|       29.37|       12.58|             98.65|
|       social|   40134|   65.09|       28.74|       12.34|             95.24|
+--------------+--------+--------+------------+------------+------------------+

+-----------+-----+---------+---------------+-----+--------+--------------------------------+---------------+-----------------+-------------------+
|category_id|views|cart_adds|purchased_lines|units|revenue |category_name                   |view_to_cart_pct|cart_to_purchase_pct|revenue_per_1k_views|
+-----------+-----+---------+---------------+-----+--------+--------------------------------+---------------+-----------------+-------------------+
|cat_002    |27516|8278     |1624           |3193 |875739.64|James LLC                       |30.08          |19.62            |31826.56           |
|cat_022    |26109|7877     |1550           |3074 |768723.02|Preston Ltd                     |30.17          |19.68            |29442.84           |
|cat_021    |31511|9499     |1854           |3728 |926467.9 |Mitchell-Kim                    |30.15          |19.52            |29401.41           |
|cat_009    |28142|8368     |1645           |3277 |826502.24|Skinner-Romero                  |29.73          |19.66            |29368.99           |
|cat_011    |25423|7626     |1519           |3021 |742053.38|Spence PLC                      |30.0           |19.92            |29188.27           |
|cat_007    |29919|9062     |1756           |3499 |873039.92|Moore, Becker and Carlson       |30.29          |19.38            |29180.12           |
|cat_000    |27407|8311     |1572           |3165 |792130.9 |Rodriguez, Figueroa and Sanchez |30.32          |18.91            |28902.5            |
|cat_020    |27091|8202     |1592           |3225 |781917.86|Brewer, Bryant and Hickman      |30.28          |19.41            |28862.64           |
|cat_023    |27019|7948     |1562           |3090 |779235.25|King-Odonnell                   |29.42          |19.65            |28840.27           |
|cat_016    |23790|7177     |1409           |2786 |680618.7 |Wright-Leblanc                  |30.17          |19.63            |28609.45           |
|cat_017    |29093|8686     |1697           |3448 |818402.64|Hancock Inc                     |29.86          |19.54            |28130.57           |
|cat_019    |22565|6801     |1293           |2591 |633219.15|Powell PLC                      |30.14          |19.01            |28062.01           |
|cat_015    |27578|8409     |1625           |3226 |773822.97|Owens-Daniel                    |30.49          |19.32            |28059.43           |
|cat_024    |30099|9035     |1686           |3360 |835988.06|Patterson, Smith and Jones      |30.02          |18.66            |27774.61           |
|cat_005    |29956|9061     |1721           |3467 |831407.06|Cameron-Parsons                 |30.25          |18.99            |27754.27           |
|cat_018    |24462|7407     |1455           |2912 |673667.06|Phillips Inc                    |30.28          |19.64            |27539.33           |
|cat_008    |29294|8894     |1629           |3280 |806409.37|Rodriguez-Graham                |30.36          |18.32            |27528.14           |
|cat_001    |27497|8160     |1542           |3124 |753182.83|Baldwin Ltd                     |29.68          |18.9             |27391.45           |
|cat_014    |28487|8502     |1632           |3282 |779652.77|Anderson, Walls and Duncan      |29.85          |19.2             |27368.72           |
|cat_012    |29460|8902     |1659           |3310 |801430.23|Carney-Santos                   |30.22          |18.64            |27204.01           |
|cat_006    |25752|7741     |1518           |3034 |695354.33|Ellis, Baker and Wright         |30.06          |19.61            |27001.95           |
|cat_013    |26649|8147     |1596           |3177 |714666.11|Smith Ltd                       |30.57          |19.59            |26817.75           |
|cat_003    |25873|7772     |1485           |2928 |687349.89|Bush Inc                        |30.04          |19.11            |26566.3            |
|cat_010    |30894|9254     |1705           |3433 |809604.06|Burns-Rodriguez                 |29.95          |18.42            |26205.87           |
|cat_004    |29723|8955     |1670           |3346 |766787.27|Kelly-Santiago                  |30.13          |18.65            |25797.78           |
+-----------+-----+---------+---------------+-----+--------+--------------------------------+---------------+-----------------+-------------------+
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week	sessions	purchase_pct	cart_pct
2026-08-10 00:00:00	9808	12.63	65.85
2026-08-17 00:00:00	15547	12.7	65.34
2026-08-24 00:00:00	15464	12.65	65.12
2026-08-31 00:00:00	15775	12.68	64.84
2026-09-07 00:00:00	15648	12.14	65.06
2026-09-14 00:00:00	15430	12.92	64.97
2026-09-21 00:00:00	15495	12.48	64.91
2026-09-28 00:00:00	15650	12.29	64.85
2026-10-05 00:00:00	15814	12.82	65.39
2026-10-12 00:00:00	15393	12.8	65.59
2026-10-19 00:00:00	15475	12.5	65.25
2026-10-26 00:00:00	15387	12.54	64.41
2026-11-02 00:00:00	15479	12.59	64.69
2026-11-09 00:00:00	3635	12.85	64.37

Comment: 200,000 sessions and 342,423 product-day rows read from HBase; funnel 200,000 → 197,509 → 130,126 → 58,215 → 25,197 (12.6%); 40,749 flagged vs 25,197 real orders; $25.5M unrealised at checkout; device, referrer, category and weekly breakdowns are all within ~1 point.